

Horses and relatives

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Perissodactyla
FAMILY	Equidae
SPECIES	7

An enduring symbol of grace and freedom, horses and their relatives (equids) are the ultimate odd-toed hoofed mammals—they have just a single toe on each foot. Equids – horses, asses (including the onager and the kiang), and zebras – have a long neck and head, and long, slender legs. They possess great stamina and can run at speed: the wild equid, the onager, can attain 43 mph (70 kph) for short periods. They are found in grasslands and deserts of Africa and Asia. Domestic horses and donkeys have been widely introduced across the world.

Anatomy

Equids are characterized by a deep chest, a mane on the neck, a tufted or long-haired tail, a solid hoof on each foot, areas of hard, thickened skin (called chestnuts) on the inside of the front legs above the knee, and mobile lips and nostrils. The eyes, which have oblong pupils, are at the sides of the head for good, all around sight (to help detect predators). Day and night vision is excellent. The ears are long and can twist to locate the sources of sounds, without having to move the body. Hearing is acute.

All species have a heavily haired coat, which is usually uniform in color in horses and asses. Zebras have striking black-and-white stripes; the function of these has had many explanations from social recognition to temperature regulation, or to create a “dazzle” effect to confuse predators.

Feeding

Horses and relatives eat mainly grass (they have a battery of hard-wearing cheek teeth for chewing), although they will also feed on desert vegetation and may browse on bark, leaves, buds, and fruit. Unlike cattle (for example), they do not ruminate but instead employ a hindgut fermentation system (see

p.211). This allows them to take in large amounts of food, which passes rapidly through the digestive tract. Quality of food is therefore less important than quantity, which means equids can survive in arid habitats. They usually rest during the heat of the day and forage in the morning, evening, and night.

Social groups

Wild horses, plains zebras, and mountain zebras live in groups consisting of mares and their young, led by a “harem” stallion, who protects and herds them. This stallion also defends the group’s territory and attempts to prevent other stallions mating with his mares. Young females may remain in the same group as their mothers or they may join a different group; young males leave at maturity and try to collect their own harem. Wild asses and Grevy’s zebras, on the other hand, have a different social organization, without long-

term associations. Breeding stallions defend large territories—up to 6 square miles (15 square km)—and these are marked by dung piles. The stallion mates with mares that range through his territory.

Equids communicate with each other by whinnying or braying, and these vocalizations vary depending on species. To assess the sexual condition of mares, stallions sniff mares’ urine. To analyze the scent in detail, they roll back their upper lip to induce the inhaled air into the Jacobson’s organ, a special pouch in the roof of the mouth. This is called the “flehmen” response.

Horses and people

The donkey was domesticated in the Middle East before 3000 BCE. The domestic horse, derived from the Eurasian wild horse, later replaced the donkey for a number of purposes – including transportation, agriculture, warfare, and recreation—although the donkey remained a popular beast of burden. Domesticated equids, including mules and hinnies, continue to serve people in many parts of the world.

Most of the wild equids are highly endangered due to habitat loss and hunting: only plains zebra and the kiang are relatively numerous.



FIGHTING FOR DOMINANCE

Conflicts between equids are common throughout the mating season. During this fight, the plains zebra stallions may bite, rear up and strike out with their front feet, and kick with their back feet. The loser is driven off, leaving the victorious stallion with control of the harem, usually between 1 and 6 mares.